

# Geometry: Ratio and Proportion (Similarity)



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## DIRECTIONS

Solve each problem. Simplify all answers. Show your work where applicable.

- 1 Simplify the ratio:

$$4 : 12$$

Answer: \_\_\_\_\_

- 2 Write as a fraction in lowest terms:

$$\frac{6}{14}$$

Answer: \_\_\_\_\_

- 3 Solve for x:

$$\frac{x}{2} = \frac{7}{7}$$

Answer: \_\_\_\_\_

- 4 Solve for x:

$$\frac{x + 14}{6} = \frac{6}{2}$$

Answer: \_\_\_\_\_

- 5 Similar triangles. Find x:

$$\frac{3}{6} = \frac{3}{x}$$

Answer: \_\_\_\_\_

- 6 Find the scale factor (small to large):

Small side = 4, Large side = 8

Answer: \_\_\_\_\_

- 7 Similar figures. Find the perimeter of the larger figure:

$$\text{Scale factor} = 4, \quad P_{\text{small}} = 24$$

Answer: \_\_\_\_\_

- 8 Angles of a triangle are in ratio. Find each angle:

Angles in ratio 2 : 4 : 3

Answer: \_\_\_\_\_

- 9 Solve the proportion (map scale):

$$\frac{1 \text{ cm}}{40 \text{ km}} = \frac{5 \text{ cm}}{x}$$

Answer: \_\_\_\_\_

- 10 Similar polygons. Find x:

$$\frac{5}{10} = \frac{6}{x}$$

Answer: \_\_\_\_\_



# Answer Key & Solutions

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## TEACHER NOTES

Reinforce cross-multiplication for solving proportions. Emphasize that similar figures share the same scale factor for all corresponding sides.

1 Simplify the ratio:

$$4:12$$

$$= 2:6$$

$GCD(4,12)=2$ . Divide both by 2.

2 Write as a fraction in lowest terms:

$$\frac{6}{14}$$

$$= \frac{3}{7}$$

Divide numerator and denominator by  $GCD = 2$ .

3 Solve for x:

$$\frac{x}{2} = \frac{7}{7}$$

$$= x = 2$$

Cross multiply:  $7x = 2 \times 7 = 14$ , so  $x = 14/7$ .

4 Solve for x:

$$\frac{x+14}{6} = \frac{6}{2}$$

$$= x = 4$$

Cross multiply:  $2(x+14) = 6 \times 6$ . So  $2x = 36 - 28 = 8$ ,  $x = 4$ .

5 Similar triangles. Find x:

$$\frac{3}{6} = \frac{3}{x}$$

$$= x = 6$$

Cross multiply:  $3x = 3 \times 6 = 18$ . So  $x = 18/3 = 6$ .

6 Find the scale factor (small to large):

Small side = 4, Large side = 8

$$= k = 2$$

Scale factor = large  $\div$  small =  $8 \div 4 = 2$ .

7 Similar figures. Find the perimeter of the larger figure:

Scale factor = 4,  $P_{\text{small}} = 24$

$$= P_{\text{large}} = 96$$

Perimeter scales with the linear scale factor:  $4 \times 24 = 96$ .

8 Angles of a triangle are in ratio. Find each angle:

Angles in ratio 2 : 4 : 3

$$= 40^\circ, 80^\circ, 60^\circ$$

Sum =  $180^\circ$ . Unit =  $180 \div (2+4+3) = 20^\circ$ . Angles:  $2 \times 20 = 40^\circ$ ,  $4 \times 20 = 80^\circ$ ,  $3 \times 20 = 60^\circ$ .

9 Solve the proportion (map scale):

$$\frac{1 \text{ cm}}{40 \text{ km}} = \frac{5 \text{ cm}}{x}$$

$$= x = 200 \text{ km}$$

Cross multiply:  $1x = 5 \times 40 = 200$ . So  $x = 200 \div 1 = 200 \text{ km}$ .

10 Similar polygons. Find x:

$$\frac{5}{10} = \frac{6}{x}$$

$$= x = 12$$

Cross multiply:  $5x = 6 \times 10 = 60$ .  $x = 60 \div 5 = 12$ .

